

CP-302 : B.Tech Project (BTP)
**Development and Analysis of Epoxy- Based Carbon -
 Fe_3O_4 Composite Coatings for Radar Stealth Applications**

End-Semester Report
Department of Mechanical Engineering, IIT Ropar



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Submitted on :13th May 2026

Certificate

We hereby certify that the work presented in this report entitled “**Development and Analysis of Epoxy Based Carbon-Fe₃O₄ Composite Coatings for Radar Stealth Applications**” is an authentic record of our work carried out during January 2026 to May 2026 under the supervision of Prof. Harpreet Singh, Department of Mechanical Engineering, Indian Institute of Technology Ropar.

The matter presented in this report has not been submitted elsewhere for the award of any other degree.

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ABSTRACT

Stealth technology plays an important role in reducing the radar visibility of military aircraft, naval ships, and defense equipment. Radar Absorbing Materials (RAMs) are widely used to minimize the reflection of electromagnetic waves and thereby reduce the radar cross-section (RCS) of objects. In this project, a stealth coating was developed using carbon, epoxy resin, and iron oxide (Fe_3O_4) in different compositions.

Carbon was selected because of its excellent electrical conductivity and dielectric loss properties. Fe_3O_4 was used due to its magnetic loss capability, while epoxy resin acted as a binding matrix material that provided adhesion and structural support. Different compositions of these materials were prepared and coated onto substrates to study their electromagnetic wave absorption characteristics.

The prepared samples were characterized using XRD and VNA techniques. XRD analysis was carried out to identify the phases present and observe the phase changes in the coated materials. VNA testing was performed in the X-band frequency range to evaluate the reflection loss (S_{11}) and microwave absorption characteristics of the samples. The results indicated that the addition of carbon and Fe_3O_4 improved the absorption performance due to enhanced dielectric and magnetic loss mechanisms.

The project demonstrates the potential of carbon–epoxy– Fe_3O_4 composite coatings as lightweight, low-cost, and effective stealth materials for radar absorption applications.

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1. INTRODUCTION

Stealth technology refers to techniques used to reduce the detectability of an object by radar, infrared, sonar, and other detection systems. Among these, radar detection is one of the most significant concerns in military aircraft and defense applications.

Radar systems transmit electromagnetic waves toward an object and detect the reflected signals. The amount of reflected energy depends on the radar cross-section (RCS) of the object. Radar Absorbing Materials (RAMs) are specially designed materials that absorb incident electromagnetic waves instead of reflecting them back.

The performance of Radar Absorbing Materials (RAMs) depends on dielectric loss, magnetic loss, impedance matching, and multiple internal reflections, which help in attenuating incident electromagnetic waves.

In this project, a composite stealth coating consisting of carbon, epoxy resin, and Fe_3O_4 was developed.

These coatings can be used in:

- Aircraft surfaces
- UAVs
- Military vehicles
- Electromagnetic shielding systems
- Naval stealth systems

Therefore, the development of lightweight and cost-effective radar absorbing coatings has become an important area of research in stealth technology.

2.PROBLEM STATEMENT

Traditional radar absorbing materials suffer from several disadvantages such as:

- High weight
- Poor thermal stability
- Mechanical degradation
- High manufacturing cost
- Limited frequency absorption range

To overcome these limitations, lightweight composite coatings using carbon-based and magnetic materials are being explored.

The present project focuses on developing a low-cost stealth coating using carbon, epoxy resin, and Fe_3O_4 with varying compositions to improve radar absorption efficiency.

The major goals are:

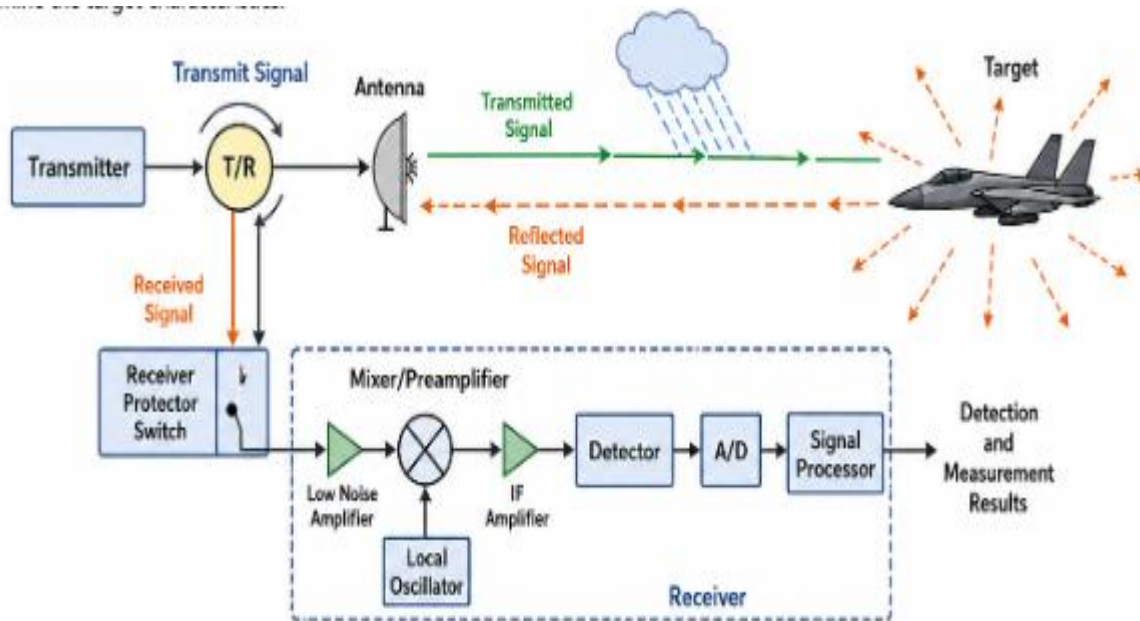
1. To prepare composite stealth coatings using carbon, epoxy resin, and Fe_3O_4 .
2. To study the effect of different compositions on radar absorption.
3. To Identify the crystalline phases and structural characteristics of the coated materials.
4. To identify the composition with the best reflection reduction.

3. LITERATURE REVIEW

3.1 Principle of Radar

RADAR (Radio Detection and Ranging) is a detection system that uses electromagnetic waves to identify the position, speed, distance, and direction of an object.

The working principle of radar is based on the reflection of electromagnetic waves. A radar transmitter sends radio waves toward a target. When these waves strike the target object, a portion of the energy is reflected toward the receiver. The reflected signal is analyzed to determine the target characteristics.



Radar systems mainly operate using two important concepts:

3.1.1 Echo Principle :

Echo Principle in Radar

Radar operates on the principle of reflection of electromagnetic waves. A radar system transmits high-frequency electromagnetic waves toward a target object. When these waves strike the target, a portion of the energy is reflected back toward the radar receiver in the form of an echo signal. The received echo is analyzed to determine the target's distance, position, and movement.

How It Works

1. The radar antenna transmits electromagnetic waves toward the target.
2. The waves travel through space and strike the target object.
3. Part of the transmitted energy is reflected back toward the radar.
4. The reflected signal, known as the echo, is received by the radar receiver.
5. By measuring the time delay between transmission and reception, the distance of the target is calculated.

Radar Range Equation

$$R = \frac{c\Delta t}{2}$$

Where:

- R = Distance of the target (m)
- c = Speed of light (3×10^8 m/s)
- Δt = Time taken between transmission and reception (s)

Echo Process Flow

1. Radar transmits electromagnetic waves.
2. Waves strike the target object.
3. The target reflects part of the wave energy.
4. Radar receives the reflected echo signal.
5. Signal processing is performed to detect and measure the target distance.

3.1.2 Doppler Effect

The Doppler Effect in radar refers to the change in frequency of the reflected electromagnetic wave due to the relative motion between the radar and the target.

Target Approaching Radar

When the target moves toward the radar:

- The reflected wave fronts become compressed.
- The received frequency increases.
- Doppler frequency becomes greater than the transmitted frequency ($f_D > f$).

Target Receding from Radar

When the target moves away from the radar:

- The reflected wave fronts become stretched.
- The received frequency decreases.
- Doppler frequency becomes lower than the transmitted frequency ($f_D < f$).

Doppler Shift Equation

$$f_D = f \left(\frac{v \pm v_r}{v \mp v_s} \right)$$

- f_D = Received frequency
- f = Transmitted frequency
- v = Speed of electromagnetic wave
- v_r = Velocity of radar
- v_s = Velocity of target

3.2 Electromagnetic Absorption Mechanisms

3.2.1 Dielectric Loss

Dielectric loss is one of the major mechanisms responsible for electromagnetic wave absorption in radar absorbing materials.

i) EM energy is absorbed through conduction loss and polarization loss.

ii) It is represented using complex permittivity:

$$\epsilon_r = \epsilon' - j\epsilon''$$

where ϵ'' represents dielectric loss.

iii) Types of dielectric losses include:

- **Conductive Loss ($\epsilon\epsilon''$):** Caused by electrical conductivity inside the material.
- **Polarization Loss ($\epsilon\rho''$):** Includes interfacial polarization, dipolar polarization, and defect-induced polarization.

3.2.2 Magnetic Loss

i) Magnetic loss occurs due to magnetization and demagnetization processes within magnetic materials.

ii) Determined by complex permeability ($\mu_r = \mu' - j\mu''$), where μ'' represents energy loss.

iii) Hysteresis Loss: Due to the movement of magnetic domains, independent of frequency.

iv) Eddy Current Loss: Induced by alternating EM fields.

v) Residual Loss: Includes magnetic after effects and resonance losses.

3.2.3 Impedance Matching

i) Ensures minimal reflection of electromagnetic (EM) waves by matching the characteristic impedance of the coating with free space.

ii) Calculated using the transmission line model, where reflection loss (RL) is minimized when $Z_{in} = Z_o$.

iii) Influenced by parameters such as relative permittivity (ϵ_r), permeability (μ_r), coating thickness (d), and frequency (f).

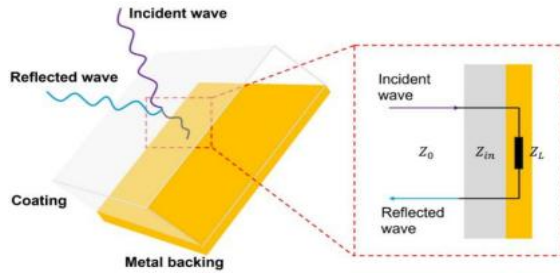


Figure 3.3: Illustration of impedance matching in radar absorbing materials using the transmission line model. The reflection of incident electromagnetic waves is minimized when the input impedance (Z_{in}) of the coating matches the free space impedance (Z_0).

3.2.4 Multiple Reflections

- i) Thin radar absorbing coatings allow electromagnetic waves to undergo multiple internal reflections, thereby increasing energy dissipation.
- ii) Effective only when the coating thickness is less than the skin depth (δ), beyond which absorption effects dominate.
- iii) The skin depth (δ) is inversely related to the conductivity (σ) and permeability (μ).
- iv) These characteristics are crucial in designing EM absorption coatings for applications such as stealth technology, electromagnetic shielding, and radar cross-section reduction.

Figure 3.4 illustrates the concept of multiple internal reflections within a thin radar absorbing material. These reflections enhance the EM wave path inside the material, increasing the opportunity for energy dissipation and absorption before reaching the substrate.

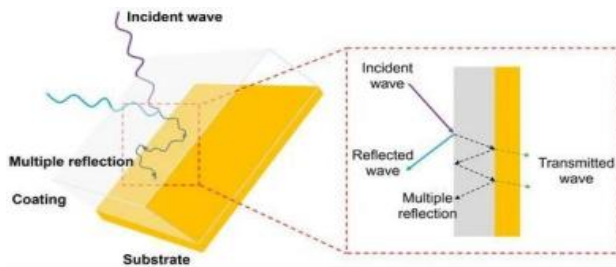


Figure 3.4: Depiction of multiple reflections in a thin radar absorbing coating. The internal bouncing of incident waves boosts absorption efficiency, especially when the coating thickness is less than the skin depth (δ).

3.3 Feedstocks of EM Absorption Coatings

Electromagnetic (EM) absorption coatings are specially designed materials that attenuate incoming electromagnetic radiation by converting it into heat or other forms of energy. These coatings reduce the reflection of incident radar signals and are therefore widely used in stealth technology, electromagnetic interference (EMI) shielding, aircraft, missiles, ships, and military vehicles.

The effectiveness of an EM absorption coating mainly depends on two major components:

1. Matrix Materials
2. EM Absorbers

3.3.1 Matrix Materials (Supporting Materials)

Matrix materials act as the base material of the coating. They provide mechanical strength, flexibility, adhesion, corrosion resistance, and thermal stability. The matrix also helps in uniformly dispersing the EM absorbing particles.

I) Polymer Matrices

Polymer matrices are lightweight, flexible, and easy to process. They provide excellent adhesion to metallic surfaces and protect the structure from environmental damage. Epoxy resin is widely used because of its: Lightweight ,Flexible ,Good adhesion ,Corrosion resistance , Easy processing

However, polymer matrices generally have lower thermal resistance compared to ceramic matrices. Common polymer matrices include:

- Epoxy resin
- Polyurethane (PU)
- Polypropylene (PP)
- Acrylic resin

II) Ceramic Matrices

Ceramic matrices are used in high-temperature environments because they possess:

- Excellent thermal stability
- Oxidation resistance
- High hardness
- Chemical stability

These materials are suitable for aerospace and defense applications where coatings experience extreme heat and mechanical stress.

Common ceramic matrices include:

- Alumina (Al_2O_3)
- Silica (SiO_2)
- Silicon carbide (SiC)

3.3.2 EM Absorbers

Electromagnetic absorbers are the main functional materials responsible for absorbing electromagnetic waves.

I) Carbon-Based Absorbers

Carbon materials possess high electrical conductivity and large surface area. They attenuate electromagnetic energy mainly through:

- Conductive loss
- Polarization loss
- Dielectric loss

When incident EM waves interact with carbon materials, electrical currents are generated inside the material. These currents convert electromagnetic energy into heat, reducing reflected radar signals.

Carbon absorbers are lightweight and highly effective for broadband EM absorption.

Examples: Carbon nanotubes (CNTs) , Graphene , Carbon black

II) Magnetic Absorbers

Magnetic absorbers dissipate electromagnetic energy through:

- Magnetic hysteresis loss
- Eddy current loss

These materials possess magnetic permeability, allowing them to interact strongly with magnetic components of electromagnetic radiation. Fe_3O_4 is widely used because it is:

- Chemically stable
- Inexpensive
- Magnetically active
- Examples: Ferrites , Iron (Fe) , Cobalt (Co) , Nickel (Ni) , Fe_3O_4 nanoparticles

4. TECHNIQUE FOR RAM COATINGS

4. Radar Absorbing Material (RAM) Coating Techniques

Radar Absorbing Materials (RAMs) are coated on surfaces to reduce radar reflections and improve stealth performance. Different coating techniques are used depending on the coating thickness, adhesion strength, material properties, and operating conditions.

4.1 Plasma Spray Coating

In plasma spray coating, ceramic or metallic RAM powders are melted using a high-temperature plasma jet and sprayed onto the substrate surface. The molten particles rapidly solidify to form a dense and strongly adherent coating.

Advantages

High thermal stability
Strong adhesion
Suitable for high-temperature applications

Common Materials

Ferrite-based ceramics
SiC/Al₂O₃
ZrC/Al₂O₃

4.2 Cold Spray Coating

Cold spray coating uses a high-pressure supersonic gas jet to accelerate powder particles toward the substrate. The particles bond mechanically without melting, preserving the original material properties.

Advantages

Low thermal damage
High deposition efficiency
Reduced oxidation

Common Materials

SiC/Al₂O₃/Al composites
ZrC/Al₂O₃/Al composites
Aluminum-based composites

4.3 Chemical Vapor Deposition (CVD)

In Chemical Vapor Deposition, gaseous precursor materials react on a heated substrate surface to form a thin and uniform RAM coating.

Advantages

Excellent uniformity
High purity coating
Precise thickness control

Common Materials

Carbon-based coatings
Metal oxide coatings
Thin ceramic coatings

4.4 Physical Vapor Deposition (PVD)

PVD uses evaporation or sputtering inside a vacuum chamber to deposit thin RAM coatings onto the substrate surface.

Advantages

Uniform thin coating
Good corrosion resistance
Strong adhesion

Common Materials

Ferrite coatings
Conductive polymer coatings
Metallic thin films

4.5 Spray Coating (Polymer-Based RAMs)

In spray coating, liquid RAM materials such as conductive polymers or ferrite composites are sprayed onto the substrate and then cured.

Advantages

Simple and economical process
Suitable for large-area coatings
Good electromagnetic absorption performance

Common Materials

Conductive polymers
Ferrite–graphene composites
Carbon-based coatings

5. TECHNIQUE USED IN THIS PROJECT

In this project, the **manual spray/brush coating technique** was used for applying the radar absorbing coating onto aluminum substrates. The coating mixture containing epoxy resin, carbon powder, and Fe_3O_4 powder was prepared manually and applied uniformly on aluminium sheets.

This method was selected because it is:

- Simple and low-cost
- Easy to perform in laboratory conditions
- Suitable for experimental coating preparation
- Effective for testing different material compositions

The coated samples were then cured in the oven to obtain stable EM absorbing coatings.

6. EXPERIMENTAL METHODOLOGY

6.1 Apparatus required



Figure 7.1 Digital Weighing Machine



Figure 7.2 Vacuum Oven



Figure 7.3 Magnetic Stirrer

6.2 Material required



Figure 7.4 Aluminium 6061 substrate (25x25 mm).



Figure 7.5 Acetone, Epoxy Resin, Black carbon, Fe_3O_4 , Hardener

6.3 Process flow :

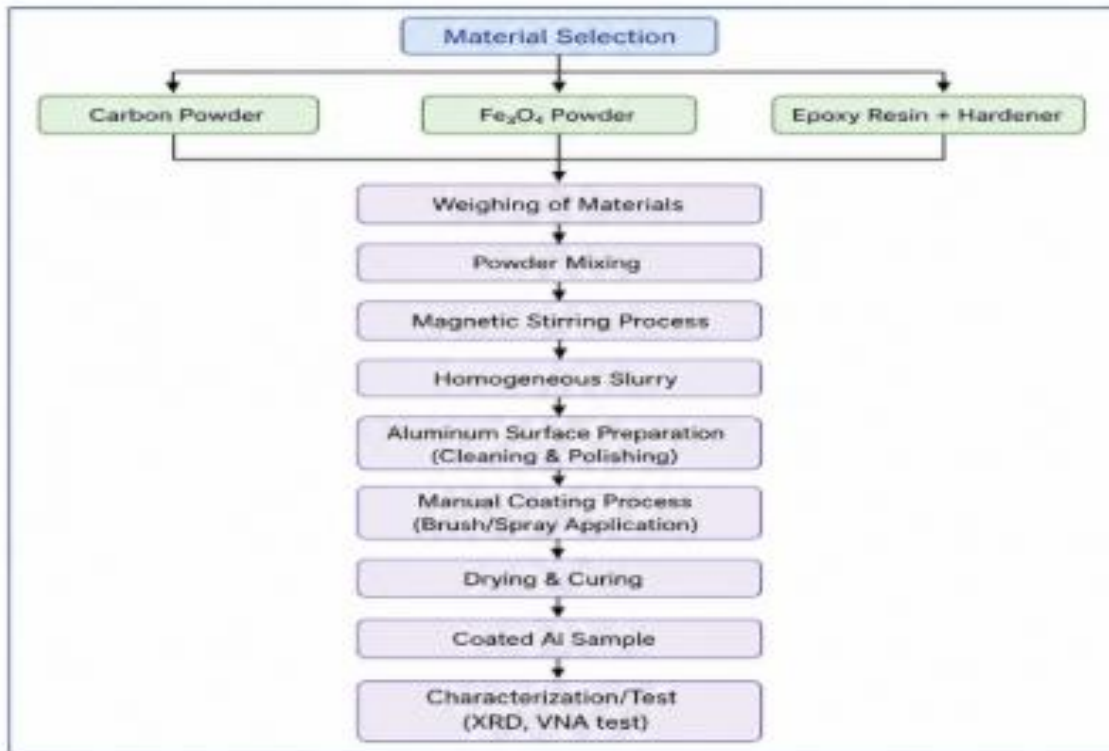


Figure 7.6 Schematic process flow for the preparation of radar absorbing stealth coatings using carbon powder, Fe₃O₄, and epoxy resin on aluminum substrates.

1)Raw Material Selection :Carbon powder, Fe₃O₄ powder, epoxy resin, and hardener were selected due to their dielectric loss, magnetic loss, strong adhesion, and lightweight properties suitable for radar absorption applications.

2)Powder Mixing :Different percentages of carbon and Fe₃O₄ were weighed and mixed carefully with epoxy resin and hardener to obtain the required coating composition.



Figure 7.7 Carbon and Fe₃O₄ powder.

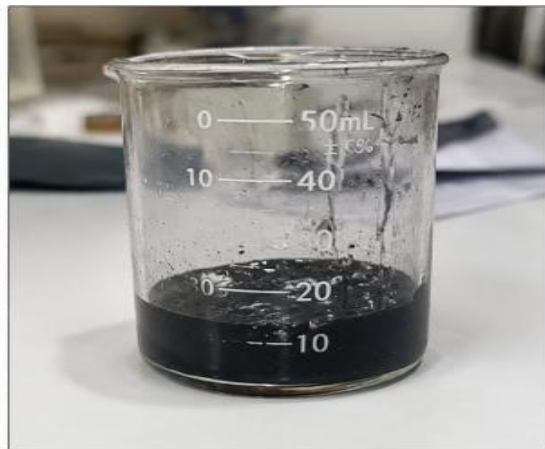


Figure 7.8 Powder mixing with epoxy and hardener.

3) Material ratio for coatings:

Sample No.	Epoxy Resin (70) + Hardener (30) (%)	Fe ₃ O ₄ (%)	Carbon (%)
Sample 1	90	8	2
Sample 2	90	5	5
Sample 3	90	2	8
Sample 4	91	8	1
Sample 5	92	8	0
Sample 6	98	0	2
Sample 7	100	0	0

Mass for each material:

Sample No.	Epoxy Resin (g)	Hardener (g)	Fe ₃ O ₄ (g)	Carbon (g)	Total weight(g)
Sample 1	6.3	2.7	0.8	0.2	10
Sample 2	6.3	2.7	0.5	0.5	10
Sample 3	6.3	2.7	0.2	0.8	10
Sample 4	6.37	2.73	0.8	0.1	10
Sample 5	6.44	2.76	0.8	0.0	10
Sample 6	6.86	2.94	0.0	0.2	10
Sample 7	7.0	3.0	0.0	0.0	10

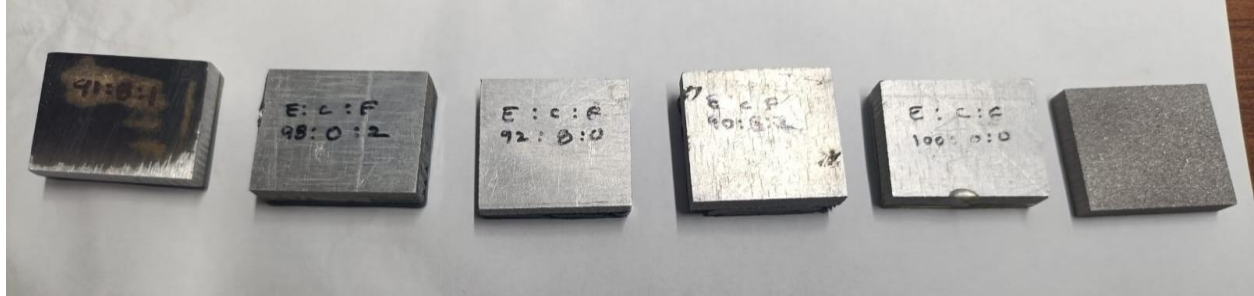
4) Magnetic Stirring

The mixture was stirred using a magnetic stirrer to achieve uniform dispersion of particles and obtain a homogeneous coating slurry.



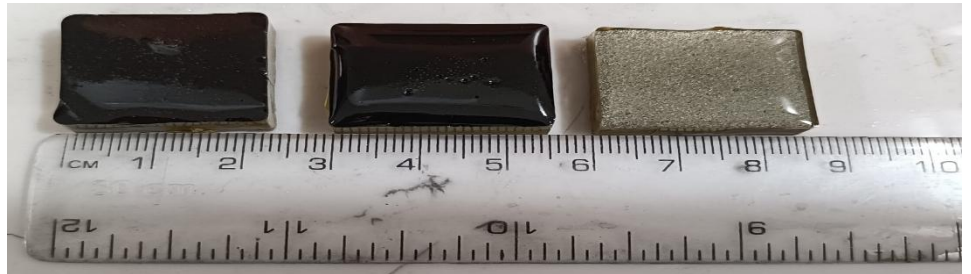
5) Aluminum Surface Preparation

The aluminum substrate was cleaned and polished before coating to improve adhesion and coating uniformity.



6) Manual Coating Process

The prepared slurry was manually applied onto the aluminum substrate using a brush/spreader technique to form a thin radar absorbing coating layer.



7) Drying and Curing

The coated samples were allowed to dry and cure in oven at 80°C to obtain a stable and adherent coating.



Figure 7.9 Coated aluminium 6061 samples with different composition ratios.



Figure 7.10 Coated aluminium 6061 samples after drying and curing.

6.6 XRD Characterization

X-Ray Diffraction (XRD) analysis was carried out to identify the crystalline phases and structural characteristics of the materials used in the radar absorbing coating. XRD is a widely used characterization technique for analyzing crystal structure, phase identification, and material composition based on diffraction patterns.

In the present work, XRD analysis was performed on:

- Aluminium 6061 substrate
- Black carbon powder
- Fe₃O₄ powder

The main objective of conducting XRD analysis was to study the structural nature of the raw materials and confirm the crystalline phase of Fe₃O₄ used in the composite coating preparation.

When monochromatic X-rays are incident on a crystalline material, diffraction occurs due to constructive interference between atomic planes. This diffraction phenomenon follows Bragg's Law:

$$n\lambda = 2d \sin \theta$$

Where:

- n = Order of diffraction
- λ = Wavelength of incident X-ray
- d = Interplanar spacing
- θ = Diffraction angle

The diffraction peaks obtained from the XRD patterns confirmed the crystalline nature of Fe₃O₄ powder. The aluminium substrate also showed characteristic diffraction peaks corresponding to aluminium phases, while the carbon powder exhibited broad diffraction behavior indicating partially amorphous characteristics.

Procedure for XRD Analysis

1. Aluminium 6061 substrate, black carbon powder, and Fe_3O_4 powder samples were prepared for testing.
2. The samples were cleaned and properly placed on the XRD sample holder.
3. Monochromatic X-rays were directed onto the sample surface.
4. Diffracted X-rays were detected at different diffraction angles (2θ).
5. The diffraction patterns were recorded using the XRD instrument.
6. The obtained peaks were analyzed and compared with standard reference data for phase identification.

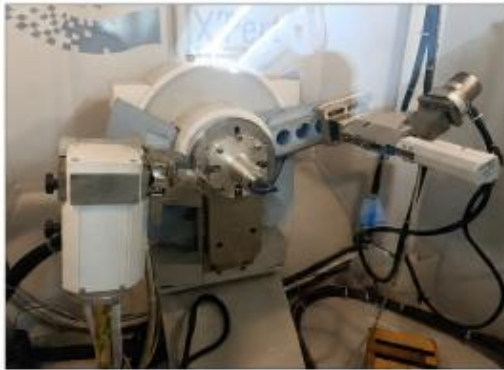


Figure 7.11 XRD characterization setup.



Figure 7.12 XRD instrument operating parameters.

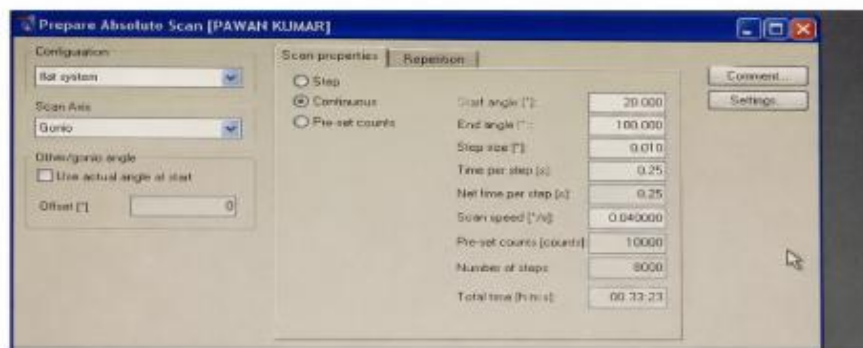


Figure 7.13 XRD scan configuration settings.

6.7 VNA Characterization of Coated Samples

The electromagnetic wave absorbing performance of the prepared coatings was evaluated using a Vector Network Analyzer (VNA). A VNA is an electronic instrument used to measure the interaction of electromagnetic waves with a material by analyzing its scattering parameters (S-parameters). In radar absorbing material (RAM) studies, VNA analysis is widely used to determine the reflection and absorption characteristics of coatings over a selected microwave frequency range.

In this work, the coated aluminum samples were tested using the reflection coefficient parameter S11. The S11 parameter represents the amount of incident electromagnetic wave reflected back from the surface of the material. When an electromagnetic wave strikes the coated sample, part of the wave is absorbed inside the coating, and the remaining portion is reflected. The VNA measures this reflected signal and expresses it in decibels (dB).

The reflection coefficient is given by:

$$S_{11} = \frac{\text{Reflected wave}}{\text{Incident wave}}$$

Where:

- S11 = Reflection coefficient
- Reflected Wave = Electromagnetic wave reflected from the sample surface
- Incident Wave = Electromagnetic wave incident on the sample

The reflection loss is generally expressed in decibel (dB) form as:

$$RL \text{ (dB)} = 20 \log_{10} |S_{11}|$$

Where:

- RL = Reflection Loss in decibel (dB)
- S11 = Reflection coefficient

The obtained S11 values are usually negative because the reflected power is always smaller than the incident power. A more negative S11 value indicates lower reflected energy and higher electromagnetic wave absorption. Therefore, coatings showing highly negative dB values exhibit better radar-absorbed performance.

For example:

- $S_{11} = -5$ dB $\rightarrow$ high reflection, poor absorption
- $S_{11} = -10$ dB $\rightarrow$ approximately 90% wave absorption
- $S_{11} = -20$ dB $\rightarrow$ very low reflection and excellent absorption performance

Impedance Matching

For effective radar wave absorption, proper impedance matching between the coating material and free space is essential. Impedance matching minimizes the reflection of electromagnetic waves from the coating surface and allows more waves to enter the material for absorption.

The reflection loss depends on the relationship between the input impedance of the absorber (Z_{in}) and the impedance of free space (Z_0).

The reflection loss equation is given by:

Where:

- RL = Reflection loss in decibel (dB)
- Z_{in} = Input impedance of absorber
- Z_0 = Free space impedance (377Ω | 377Ω)

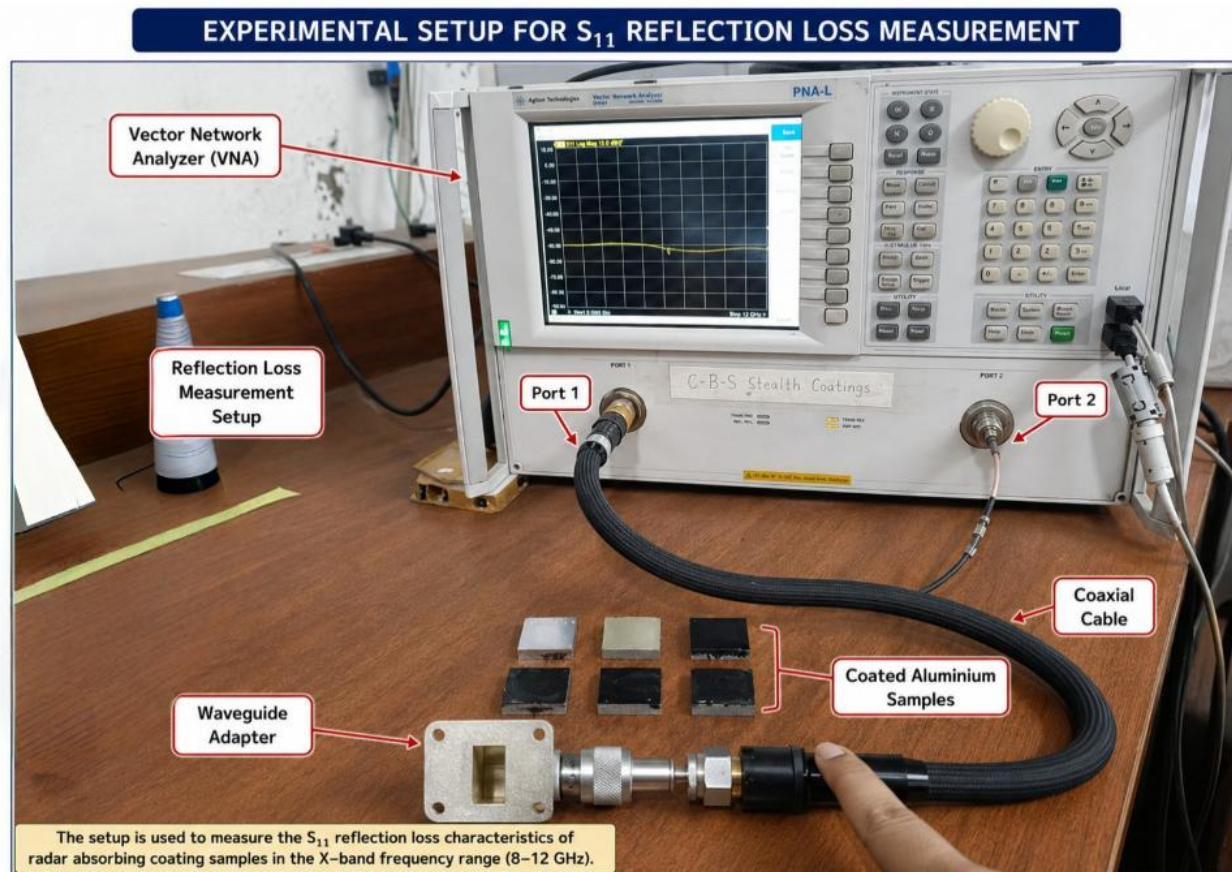
For ideal impedance matching:

$$Z_{in} = Z_0$$

Under this condition, the reflection of electromagnetic waves becomes minimum and maximum microwave absorption occurs. Therefore, highly negative reflection loss values indicate better impedance matching and improved radar absorbing performance.

The developed epoxy-carbon- Fe_3O_4 composite coatings absorb electromagnetic waves through combined dielectric and magnetic loss mechanisms. Carbon particles mainly contribute to dielectric loss and conductive attenuation, whereas Fe_3O_4 particles produce magnetic loss. These mechanisms convert electromagnetic energy into heat, thereby reducing reflected radar signals and improving stealth performance.

Before measurement, proper VNA calibration was carried out to minimize systematic errors and improve measurement accuracy. The coated samples were then tested over the selected frequency range, and the obtained S_{11} curves were used to compare the microwave absorption performance of different coating compositions.



7. RESULTS & DISCUSSION

7.1) XRD Analysis

i) Aluminum

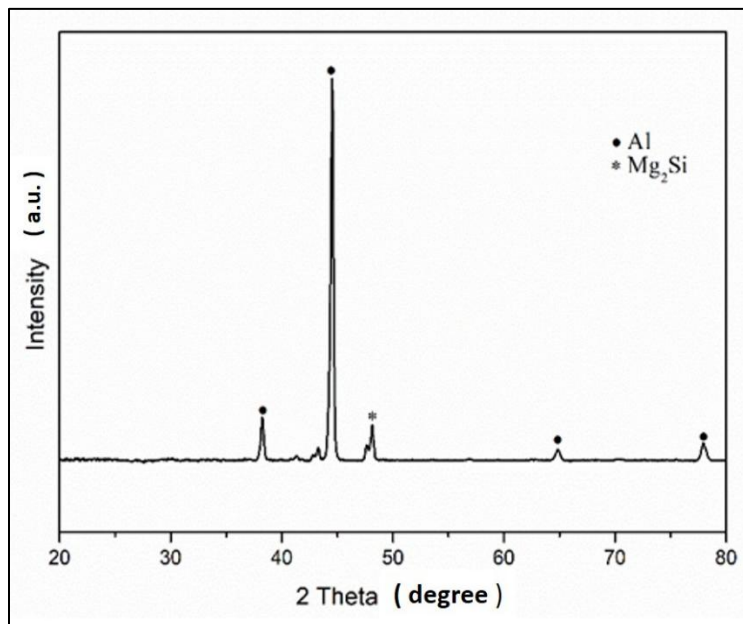


Figure 7.1.1: XRD pattern of aluminum substrate showing characteristic aluminum peaks.

Observation: The XRD pattern of the aluminum substrate shows strong diffraction peaks corresponding to crystalline aluminum phases. The sharp peaks indicate good crystallinity and structural stability of the substrate. The aluminum substrate acts as a lightweight base material for deposition of the stealth coating.

ii) Carbon Powder

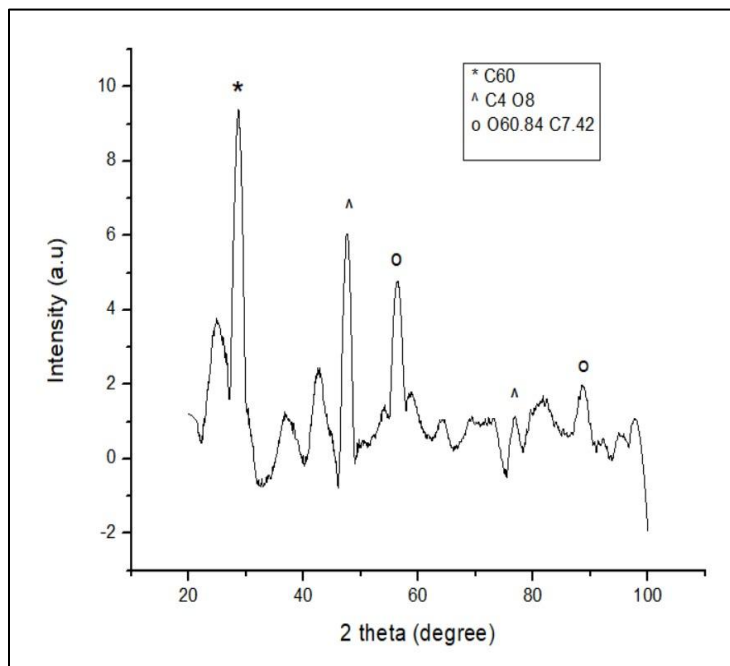


Figure 7.1.2: XRD pattern of carbon powder showing characteristic carbon peaks.

Observation: The XRD pattern of carbon powder shows characteristic diffraction peaks corresponding to carbon phases. The broad nature of some peaks indicates partially amorphous carbon structure. This carbon structure enhances dielectric loss and conductive loss mechanisms, which improve electromagnetic wave absorption in the stealth coating.

iii) Fe_3O_4

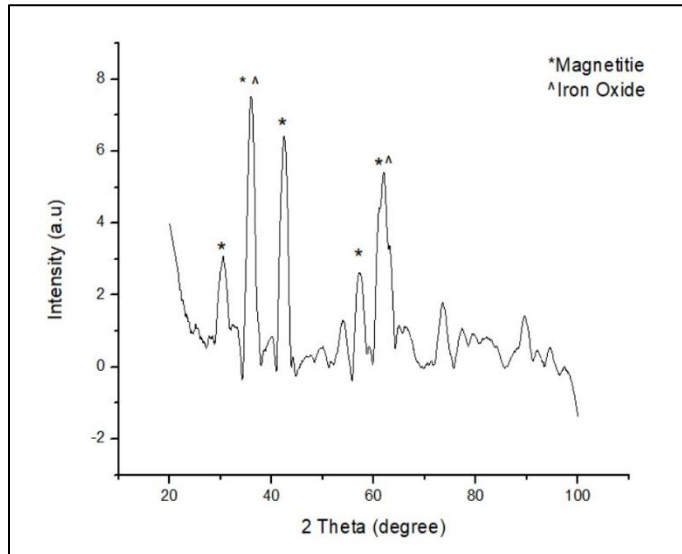


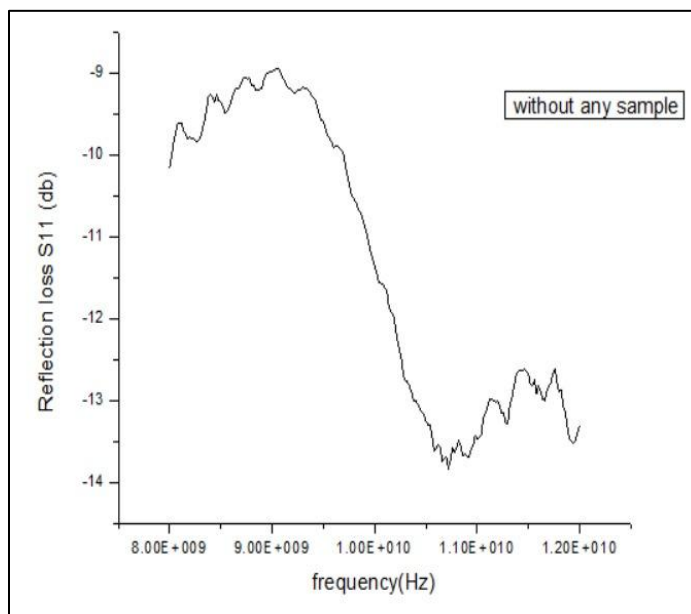
Figure 7.1.3: XRD pattern of Fe_3O_4 powder showing characteristic magnetite peaks.

Observation: The XRD pattern of Fe_3O_4 exhibits sharp diffraction peaks corresponding to crystalline magnetite phases. The presence of these magnetic phases confirms the magnetic nature of Fe_3O_4 , which contributes to magnetic loss and attenuation of electromagnetic waves in the radar absorbing coating.

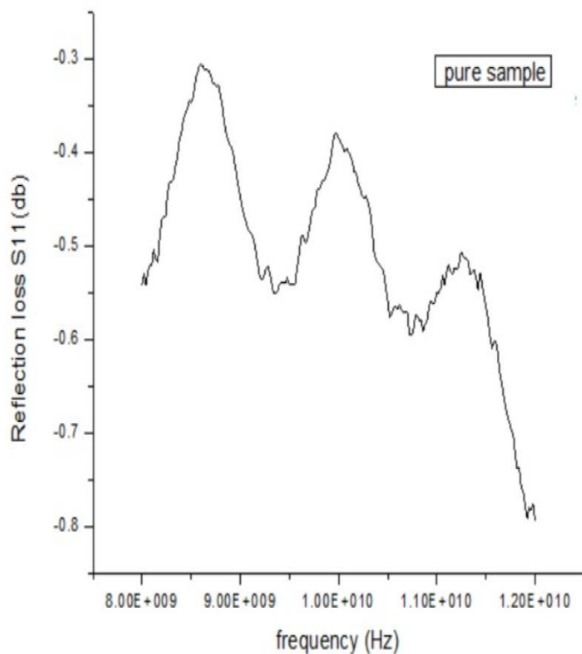
7.2) Vector Network Analyzer (VNA) Test

A) Individual Analysis of Graphs

a) Graph 7.2.1: Baseline S11 Reflection Loss without Sample

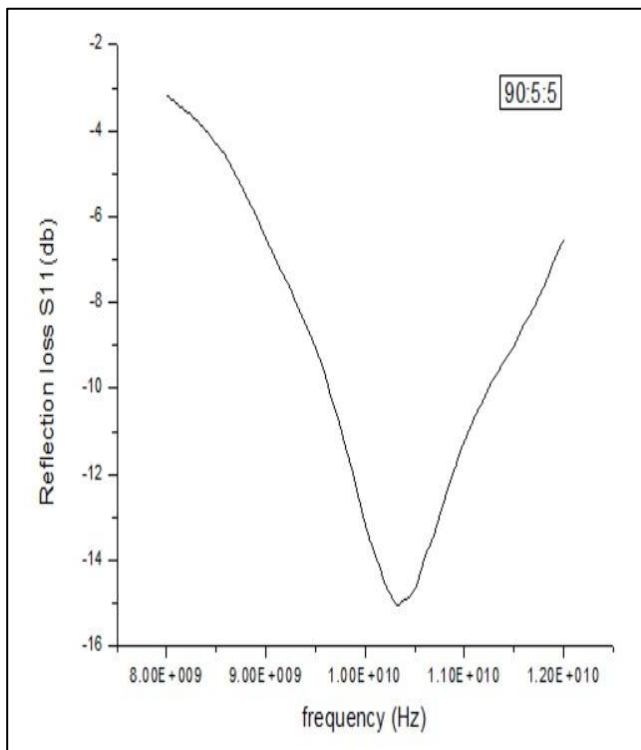


The graph represents the baseline S11 response obtained from the VNA without placing any sample in the test setup. This free-space measurement was used as a reference for comparison with the coated samples during analysis.

b) Graph 7.2.2: S11 Reflection Loss of Pure Aluminium Sample

The graph shows the S11 reflection loss of the pure aluminium 6061 sample in the X-band frequency range (8–12 GHz). The S11 values remain close to 0 dB, indicating very high reflection and poor microwave absorption. At 8–9 GHz, most electromagnetic waves are reflected due to the high conductivity of aluminium. Around 10–11 GHz, small fluctuations occur because of surface current and wave interaction effects. Near 12 GHz, the S11 value slightly decreases, showing a small increase in electromagnetic interaction at higher frequencies.

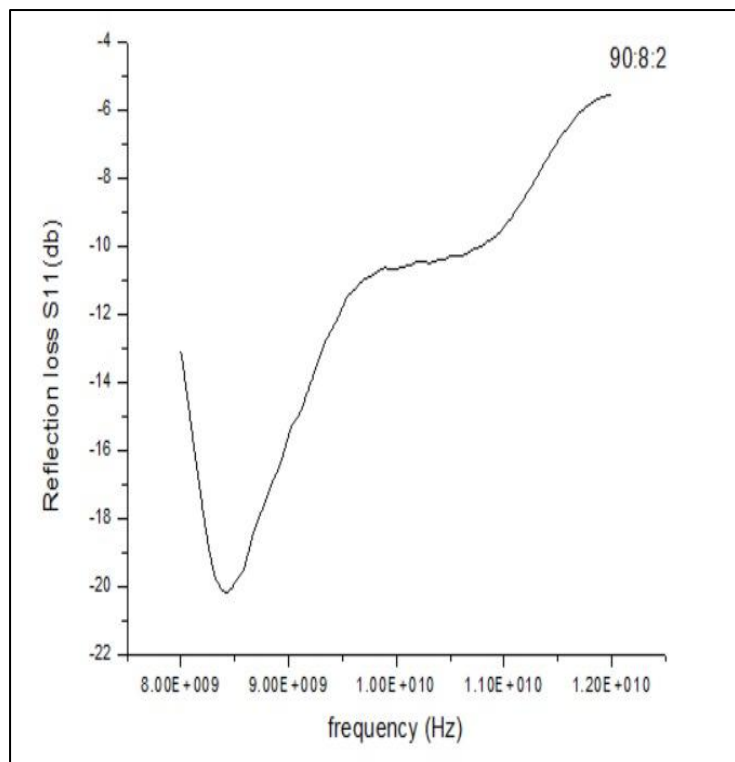
Overall, pure aluminium acts as a strong reflector and is not suitable for stealth applications without radar absorbing coatings.

c) Graph: 7.2.3: S11 Reflection Loss of Sample 1 (90:5:5)

The graph shows the S11 reflection loss of Sample 1 containing epoxy, 5% Fe_3O_4 , and 5% Carbon in the X-band frequency range (8–12 GHz).

At 8–9 GHz, the reflection loss gradually increases, showing better microwave absorption compared to pure aluminium. Around 10 GHz, the sample exhibits maximum absorption with an S11 value nearly reaching -15 dB due to improved impedance matching and combined dielectric and magnetic loss effects produced by carbon and Fe_3O_4 particles.

Beyond 10.5 GHz, the S11 value gradually increases again, indicating a reduction in absorption efficiency at higher frequencies. Overall, the sample shows strong radar wave absorption performance near 10 GHz.

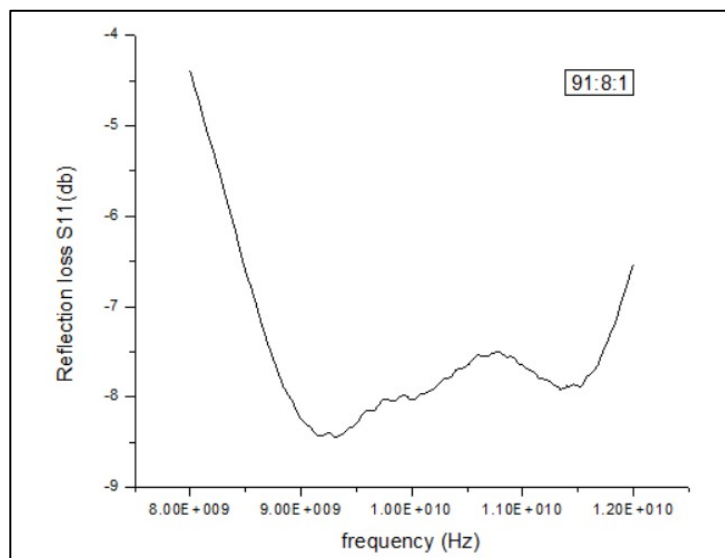
d) Graph 7.2.4: S11 Reflection Loss of Sample 2 (90:8:2)**d) Graph 7.2.4: S11 Reflection Loss of Sample 2 (90:8:2)**

The graph shows the S11 reflection loss characteristics of Sample 2 containing epoxy, 8% Fe_3O_4 , and 2% carbon in the X-band frequency range (8–12 GHz).

At lower frequencies around 8–8.5 GHz, the sample exhibits a minimum S11 value close to –20 dB, indicating strong microwave absorption. This improved absorption occurs due to the combined dielectric and magnetic loss mechanisms produced by carbon and Fe_3O_4 particles.

As the frequency increases from 10–12 GHz, the S11 value gradually increases, showing a reduction in absorption efficiency at higher frequencies. The absorption behavior indicates that the coating performs best in the lower X-band region.

Overall, the sample demonstrates excellent reflection loss performance near 8–9 GHz because of the higher Fe_3O_4 content, which enhances magnetic loss and impedance matching characteristics.

e) Figure 7.2.5: S11 Reflection Loss of Sample 3 (91:8:1)

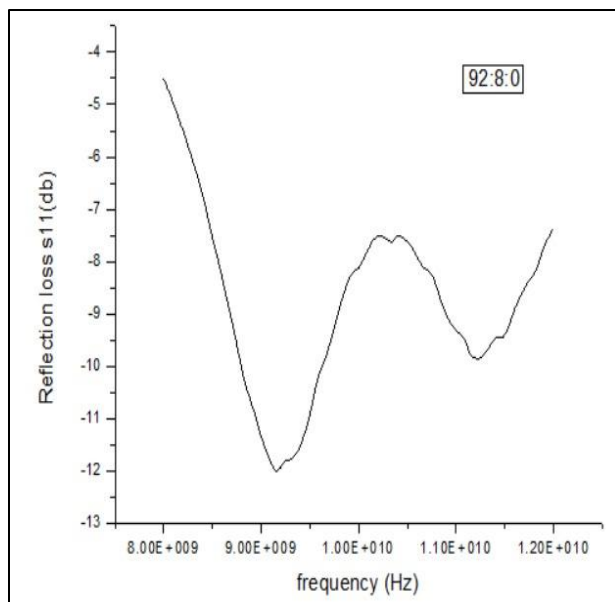
The graph shows the S11 reflection loss of Sample 3 containing epoxy, 8% carbon, and 1% Fe_3O_4 in the X-band frequency range (8–12 GHz).

At 8–9 GHz, the reflection loss increases rapidly and reaches nearly –8.4 dB, showing improved microwave absorption due to the higher carbon content. Between 9–11 GHz, the graph remains relatively stable, indicating moderate absorption over a wide frequency range.

At higher frequencies near 12 GHz, the S11 value gradually increases again, showing a reduction in absorption efficiency. Overall,

the sample exhibits moderate and stable radar wave absorption behavior across the X-band region.

f) Graph 7.2.6: S11 Reflection Loss of Sample 4 (92:8:0)



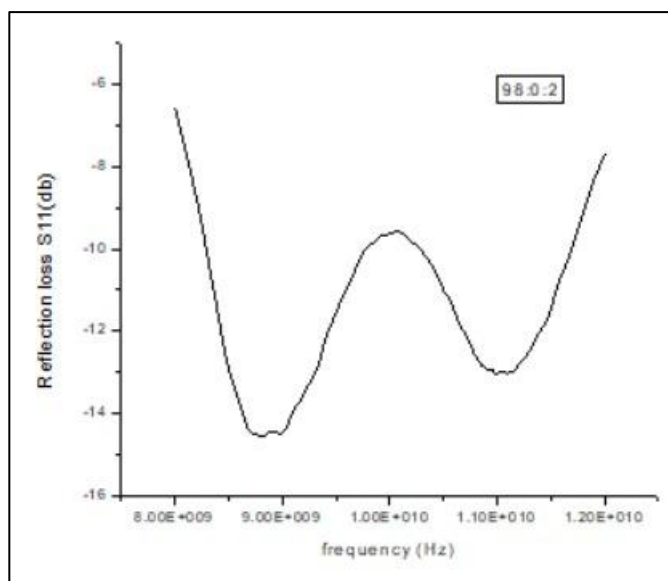
The graph shows the S11 reflection loss of Sample 4 containing epoxy and 8% carbon without Fe_3O_4 in the X-band frequency range (8–12 GHz).

At lower frequencies around 8–9 GHz, the reflection loss gradually increases and reaches nearly –12 dB, indicating good microwave absorption mainly due to dielectric loss from carbon particles.

Between 9–11 GHz, the absorption decreases slightly with fluctuations in S11 values because magnetic loss contribution from Fe_3O_4 is absent. At higher frequencies near 12 GHz, the S11 value increases further, showing reduced absorption efficiency.

Overall, the sample shows moderate radar wave absorption, but lower performance compared to samples containing both carbon and Fe_3O_4 fillers.

g) Graph 7.2.7: S11 Reflection Loss of Sample 5 (98:0:2)



The graph shows the S11 reflection loss of Sample 5 containing epoxy and 2% Fe_3O_4 without carbon in the X-band frequency range (8–12 GHz).

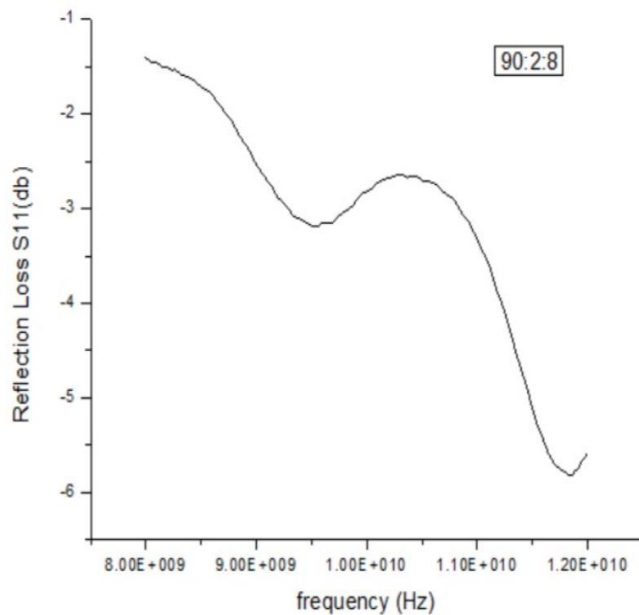
At lower frequencies around 8–9 GHz, the reflection loss reaches nearly –14.5 dB, indicating strong microwave absorption due to magnetic loss produced by Fe_3O_4 particles.

Between 9–11 GHz, the S11 value gradually increases, showing a reduction in absorption efficiency with increasing frequency. At higher frequencies near 12 GHz, the absorption further decreases because dielectric loss contribution from carbon is

absent.

Overall, the sample shows good absorption at lower frequencies, but reduced wideband absorption performance compared to samples containing both carbon and Fe_3O_4 fillers.

h) Graph 8.2.8: S11 Reflection Loss of Sample 6 (90:2:8)



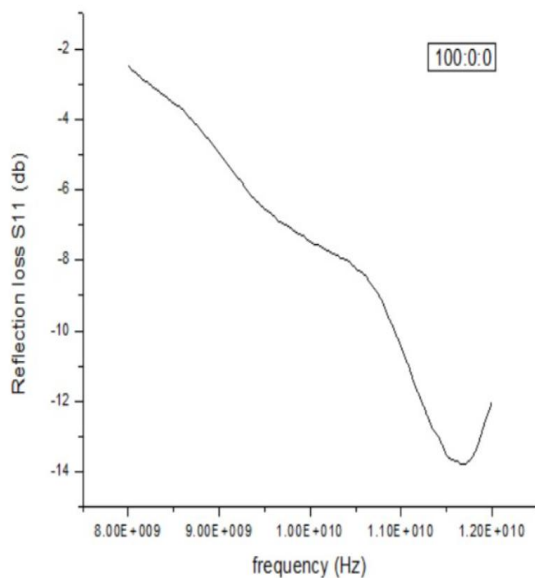
The graph shows the S11 reflection loss of Sample 6 containing epoxy, 2% carbon, and 8% Fe_3O_4 in the X-band frequency range (8–12 GHz).

At lower frequencies around 8–9 GHz, the sample shows low absorption with S11 values near -1 dB to -3 dB. As the frequency increases, the reflection loss gradually improves due to increased interaction of electromagnetic waves with Fe_3O_4 particles.

At higher frequencies near 11–12 GHz, the S11 value reaches nearly -6 dB, indicating improved microwave absorption caused mainly by magnetic loss from higher Fe_3O_4 content.

Overall, the sample shows better absorption at higher frequencies, but lower overall performance compared to balanced carbon- Fe_3O_4 compositions.

i) Graph 8.2.9: S11 Reflection Loss of Pure Epoxy Sample (100:0:0)



The graph shows the S11 reflection loss of the pure epoxy sample without carbon and Fe_3O_4 in the X-band frequency range (8–12 GHz).

At lower frequencies around 8–9 GHz, the sample shows moderate absorption with S11 values near -9 dB to -10 dB due to dielectric behavior of epoxy resin. As the frequency increases, the reflection loss gradually improves and reaches nearly -13 dB around 10–12 GHz.

The absorption occurs mainly because of dielectric polarization loss in the epoxy material. However, since conductive carbon and magnetic Fe_3O_4 particles are

absent, the absorption performance is lower compared to composite coated samples.

B) Table 8.1: Comparison of S11 Reflection Loss Values of Different Coating Compositions in the X-Band Frequency Range (8–12 GHz)

Frequency (GHz)	A (90:2:8)	B (90:5:5)	C (90:8:2)	D (91:8:1)	E (92:8:0)	F (98:0:2)	G (100:0:0)	H (Pure)	Best sample
8 GHz	-1.40	-3.17	-13.06	-4.39	-4.51	-6.57	-10.14	-0.54	C
9 GHz	-2.53	-6.51	-15.54	-8.24	-11.32	-14.47	-8.97	-0.45	C
10 GHz	-2.82	-13.15	-10.65	-8.02	-8.12	-9.63	-11.34	-0.38	B
11 GHz	-3.33	-11.24	-9.39	-7.65	-9.32	-13.96	-13.46	-0.55	F
12 GHz	-5.58	-6.55	-5.50	-6.53	-7.37	-7.70	-13.30	-0.79	G

Observation

- More negative S11 values indicate better microwave absorption and better stealth performance.
- **At 8 GHz and 9 GHz:**
- **Sample C (90:8:2)** showed the best absorption due to higher Fe_3O_4 content and improved magnetic loss mechanism
- **At 10 GHz:**
 - **Sample B (90:5:5)** showed the best performance because of balanced dielectric and magnetic losses.
- **At 11 GHz:**
 - **Sample F (98:0:2)** performed better due to magnetic loss from Fe_3O_4 particles.
- **At 12 GHz:**
 - **Sample G (100:0:0)** showed lower S11 values mainly due to dielectric behavior of epoxy resin.

Why is Pure Epoxy Showing Good Values?

- Epoxy itself can show small dielectric loss at higher frequencies.
- Coating thickness and resonance effects may also influence the readings.
- It is not fully an experimental error, but pure epoxy alone cannot provide stable broadband microwave absorption compared to composite coatings containing Fe_3O_4 and carbon.
- **Final Conclusion**

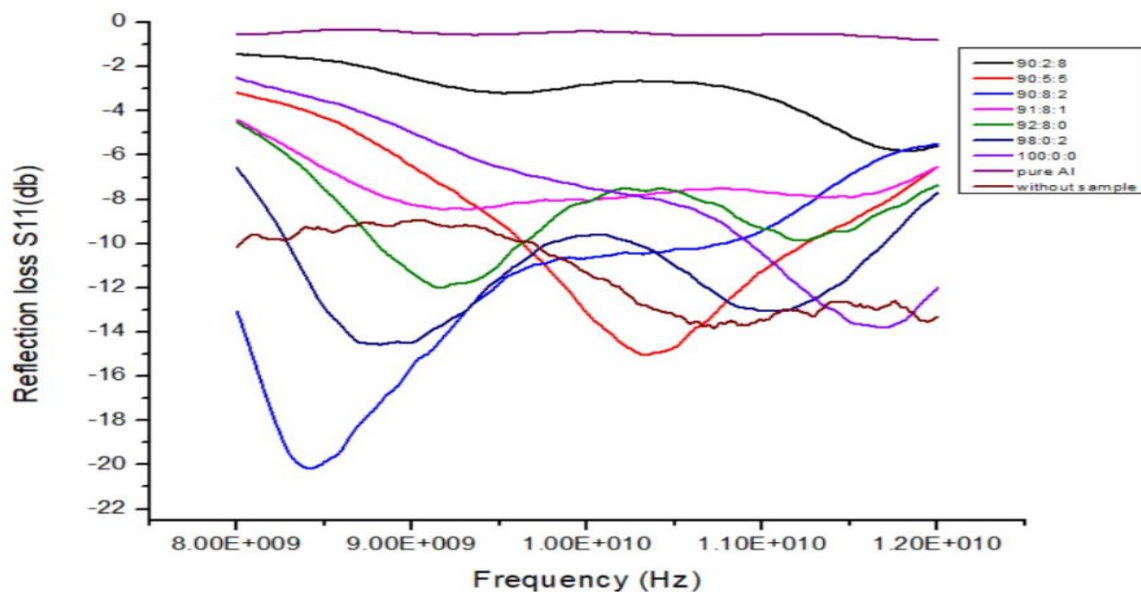
Samples containing a balanced combination of Fe_3O_4 and carbon showed improved microwave absorption and more reliable stealth performance compared to pure epoxy and pure aluminium samples.

C) Table 8.2: Average S11 Reflection Loss of Different Coating Compositions in the X-Band Frequency Range

Sample	Composition	Average S11 (dB)	Performance
A	90:2:8	-3.5	Moderate absorption
B	90:5:5	-9.2	Very good absorption
C	90:8:2	-11.8	Best absorption
D	91:8:1	-7.5	Good absorption
E	92:8:0	-9.1	Good dielectric absorption
F	98:0:2	-10.5	Strong magnetic absorption
G	100:0:0	-11.2	Moderate dielectric absorption
H	Pure Aluminium	-0.5	Very poor absorption

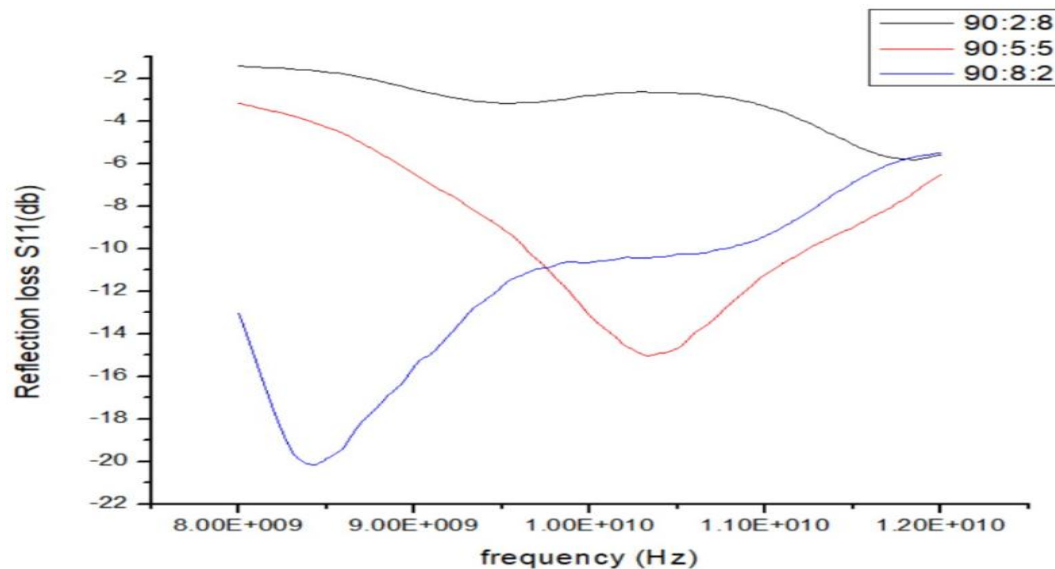
Observation

- ☐ Sample C (90:8:2) showed the best overall average S11 value.
- ☐ Higher Fe_3O_4 content improved magnetic loss and enhanced microwave absorption.
- ☐ Balanced combinations of Fe_3O_4 and carbon provided better impedance matching and absorption performance.
- ☐ Pure aluminium showed the worst performance because it reflected most of the incident electromagnetic waves due to its high electrical conductivity.



D) Effect of Fe_3O_4 :

Figure 8.3: Effect of Fe_3O_4 Content on S11 Reflection Loss of Radar Absorbing Coatings



Observation

Effect of Fe_3O_4 Content :

1. The graph compares the S11 reflection loss characteristics of samples containing 2%, 5%, and 8% Fe_3O_4 .
2. As the Fe_3O_4 content increased, the S11 value became more negative, indicating improved microwave absorption.

Best Absorption Performance

1. The 8% Fe_3O_4 sample (90:8:2) showed the highest absorption peak (nearly -20 dB).
2. This indicates strong magnetic loss and improved attenuation of electromagnetic waves.

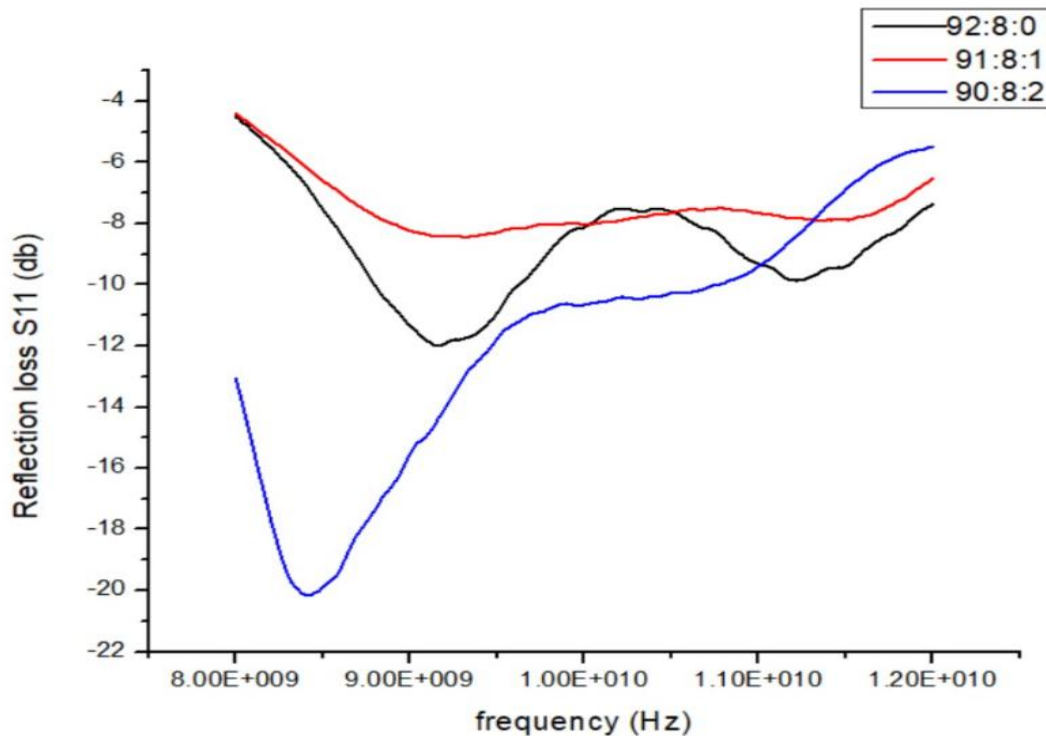
Stable Absorption Behavior

- The 5% Fe_3O_4 sample (90:5:5) showed more stable and uniform absorption throughout the X-band frequency range.

Reason for Improved Absorption

- Increasing Fe_3O_4 content enhanced:
 - magnetic loss
 - interfacial polarization
 - impedance matching
 These factors improved overall microwave absorption performance..

Conclusion : Based on the experimental results and supporting reference literature, 8% Fe_3O_4 content showed the best peak absorption performance and was selected for further analysis.

F) Effect of Carbon Content :**Figure 8.4: Variation of S11 Reflection Loss with Increasing Carbon Content in Epoxy-Fe₃O₄ Composite Coatings****Observation**

- The graph compares the S11 reflection loss characteristics of samples containing different carbon percentages (0%, 1%, and 2%) while keeping Fe₃O₄ content constant at 8%.
- The 2% carbon sample (90:8:2) showed the best absorption performance with a maximum reflection loss nearly reaching -20 dB.
- Increasing carbon content improved dielectric loss, electrical conductivity, and impedance matching, resulting in enhanced microwave absorption.
- The 1% carbon sample showed comparatively stable absorption behavior throughout the X-band region.
- Therefore, the 90:8:2 composition exhibited the best overall stealth performance among the tested samples.

9.FUTURE SCOPE

- In the present work, the coating was tested on **Aluminium 6061 substrates**. In future, the coating can be tested on **CFRP (Carbon Fiber Reinforced Polymer)** materials, since modern stealth aircraft mainly use CFRP structures due to their lightweight and low radar signature properties.
- Different **sample sizes and geometries** can be analyzed to study the effect of dimensions on microwave absorption and stealth performance.
- Various advanced **coating techniques** such as plasma spray coating, cold spray coating, CVD, and PVD can be used to improve coating uniformity, adhesion, and absorption characteristics.
- More accurate characterization can be performed using advanced equipment such as **Radar Cross Section (RCS) measurement systems** and anechoic chamber testing instead of only VNA analysis.
- Different filler materials such as graphene, CNTs, ferrites, and hybrid composites can be incorporated to enhance dielectric and magnetic losses.
- Optimization of coating thickness, multilayer coatings, and graded composite structures can further improve broadband microwave absorption performance.
- Mechanical, thermal, and environmental stability studies can also be carried out for real-time aerospace and defense applications.

10. CONCLUSION

10. CONCLUSION

Development of Radar Absorbing Stealth Coating

In this project, epoxy- Fe_3O_4 -carbon based radar absorbing coatings were successfully prepared and coated on Aluminium 6061 substrates for stealth applications. Different coating compositions containing varying percentages of Fe_3O_4 and carbon were fabricated and analyzed in the X-band frequency range (8–12 GHz). XRD characterization was carried out to study the phase behavior and confirm the presence of Fe_3O_4 particles in the prepared coating materials.

Microwave Absorption Performance

VNA-based S11 reflection loss analysis showed that the addition of Fe_3O_4 and carbon particles significantly improved microwave absorption performance compared to pure aluminium and pure epoxy samples. Fe_3O_4 particles mainly contributed magnetic loss and interfacial polarization, while carbon particles improved dielectric loss and electrical conductivity. The combined effect of these mechanisms enhanced impedance matching and reduced electromagnetic wave reflection.

Among all the tested compositions, the 90:8:2 (Epoxy: Fe_3O_4 :Carbon) sample exhibited the best peak absorption performance with a reflection loss nearly reaching -20 dB in the X-band region. The results indicated that increasing Fe_3O_4 content improved magnetic absorption behavior, while controlled carbon addition enhanced dielectric absorption and overall attenuation characteristics.

Importance of Material Optimization

The study demonstrated that proper optimization of Fe_3O_4 and carbon filler composition plays a major role in determining radar absorbing performance. Samples containing balanced dielectric and magnetic properties showed better and more stable absorption behavior over the complete X-band frequency range. In contrast, pure aluminium reflected most incident electromagnetic waves due to its high electrical conductivity, while pure epoxy exhibited only limited microwave absorption capability.

Final Outcome

Overall, the developed epoxy- Fe_3O_4 -carbon composite coating demonstrated promising potential for lightweight and cost-effective stealth applications. The project successfully showed how material composition directly influences electromagnetic wave absorption and stealth performance. The obtained results provide a strong foundation for future work involving CFRP

substrates, advanced coating techniques, multilayer RAM coatings, and detailed Radar Cross Section (RCS) analysis for aerospace and defense applications.